

planet are expected to produce seasonal temperature variations of around 30 K.

The planetary orbit is aligned with the stellar orbit to within 0.4° . This extreme coplanarity suggests that the planet was formed along with the stars, within a circumbinary protoplanetary disk, as opposed to being captured from another system. Planetesimal formation around an eccentric binary is a theoretical challenge, because of the large collision velocities of particles that are stirred by the stellar binary (24), although the detection of debris disks around close binaries has been interpreted as dust produced by colliding planetesimals (25). Subsequent stages of planet formation around binaries have been studied theoretically, both for terrestrial planets (26) and gas giants (27), but these and other theoretical studies (28) have lacked a well-specified circumbinary planetary system that could allow such a refinement of models.

Finally, the stars themselves are worthy of attention, independently of the planet. It is rare to measure the masses and radii of such small stars with such high precision, using geometrical and dynamical methods independent of stellar evolutionary models. In particular, star B, with only 20% the mass of the Sun, is the smallest main-sequence star for which such precise mass and radius data are available (29) (Fig. 3). The mass ratio of 0.29 is also among the smallest known for binaries involving fully convective stars at the low-mass end of the main sequence (28). With well-characterized low-mass stars, in

addition to a transiting circumbinary planet, this makes Kepler-16 a treasure for both exoplanetary and stellar astrophysical investigations.

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Supporting Online Material

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Tables S1 to S5
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REPORTS

Unclicking the Click: Mechanically Facilitated 1,3-Dipolar Cycloreversions

Johnathan N. Brantley, Kelly M. Wiggins, Christopher W. Bielawski*

The specific targeting of covalent bonds in a local, anisotropic fashion using mechanical methods offers useful opportunities to direct chemical reactivity down otherwise prohibitive pathways. Here, we report that embedding the highly inert 1,2,3-triazole moiety (which is often prepared using the canonical "click" coupling of azides and alkynes) within a poly(methyl acrylate) chain renders it susceptible to ultrasound-induced cycloreversion, as confirmed by comprehensive spectroscopic and chemical analyses. Such reactivity offers the opportunity to develop triazoles as mechanically labile protecting groups or for use in readily accessible materials that respond to mechanical force.

The 1,3-dipolar cycloaddition of azide and alkyne moieties (1, 2), which allows access to a variety of substituted triazoles, is included under the umbrella of "click" chemistry. This reaction has found broad applicability over

the past decade because it exhibits rapid kinetics under mild conditions, high functional group and solvent tolerance, and good atom economy, and it has a propensity to generate relatively chemically inert and thermally stable products. In addition to finding compelling use in molecular and polymer functionalizations, this coupling motif has been applied to robust, chemically orthogonal ligations for the study of biological systems (3–11). However, a consequence of the high kinetic stability of these

triazole products is that simple chemical or thermal treatments capable of cleanly reverting the coupling reactions into their constituent azides and alkynes are unknown. We envisioned that mechanical force could be used to surmount the otherwise inaccessible barrier to triazole cycloreversion; that is, under the appropriate mechanical stress, triazoles might not retain their structural integrity. Such a retro-cycloaddition would also indicate that triazoles are not necessarily orthogonal to chemical transformations and would provide a method by which reactive azide or alkyne intermediates could be selectively unmasked to effect desired transformations (Scheme 1).

Recent advances in mechanochemistry (12)—wherein exogenous forces can be directed to mechanophores, or small molecules possessing mechanically labile bonds, through the judicious attachment of polymer chains—have demonstrated that formally disallowed pericyclic reactions and thermally inaccessible isomerizations can be readily induced through site-specific mechanical activation (13–18). Such forces can be harnessed through the application of ultrasound to polymer solutions, whereby cavitation induces velocity gradients and attendant stress to the

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solvated polymer chains (12). Mechanical forces are presumed to promote otherwise prohibited reactions through ground-state destabilization of the reactants (as a result of changes in molecular geometry) or the stabilization of reactive intermediates at or near the transition state of the reaction coordinate (19, 20). Based on this foundation, we hypothesized that triazoles, although remarkably inert toward chemical and thermal perturbation, could be coerced to undergo cycloreversion through the strategic application of mechanical force.

To assess the mechanophoricity of the triazole moiety, diol **1** was condensed with 2-bromoisobutyl

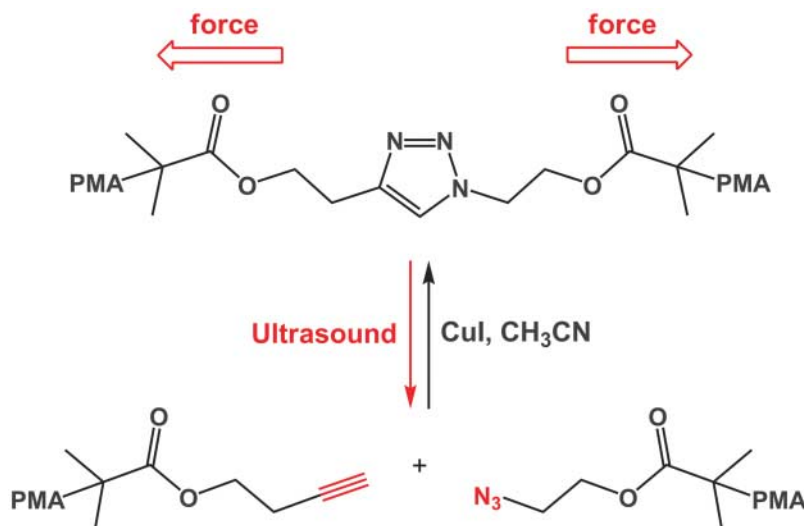
bromide to generate the bifunctional initiator **3** (Scheme 2). Subsequent Cu-mediated single-electron transfer living radical polymerization (SET LRP) of methyl acrylate from **3** resulted in a triazole-centered poly(methyl acrylate) (TriP_{M_n} , where M_n refers to the number average molecular weight of the polymer) (Scheme 2). Analysis by gel permeation chromatography (GPC) revealed that the material (TriP_{96}) displayed an M_n of 96 kD and a polydispersity index (PDI) of 1.3 (21). Subjecting an acetonitrile (CH_3CN) solution of TriP_{96} to ultrasound for 2 hours in a Suslick cell (22) at 0°C resulted in a decrease in the polymer's M_n from 96 to 48 kD

(Fig. 1A, black and red lines, respectively). This change was consistent with chain scission near the center of the polymer, which indicated the potential activation (i.e., cycloreversion) of the centrally located mechanophore. Additionally, the infrared (IR) spectrum of the postsonicated polymer revealed diagnostic signals consistent with the presence of azide (2133 cm^{-1}) and terminal alkyne (3021 and 2039 cm^{-1}) functionalities that were not present in the IR spectrum of the pre-sonicated material (Fig. 1B).

To further demonstrate that the result of mechanical scission was the expected azide and alkyne products of triazole cycloreversion, we sought to selectively label these moieties with diagnostic chromophores possessing complementary functionalities. Pyrene derivatives exhibit distinct absorbances in the ultraviolet-visible (UV-vis) spectrum; thus, 1-azidopyrene was chosen to selectively label the mechanically liberated alkyne with a chromophore through the [3+2] cycloaddition process of classic click chemistry (Scheme 2). The postsonicated TriP_{96} was dissolved in CH_3CN , and 1-azidopyrene was cycloadded to the free alkyne using CuI. GPC with concomitant UV-vis detection revealed that the labeled polymer had an increased absorbance at $\lambda = 350\text{ nm}$ (a λ_{max} associated with pyrene) relative to the starting material at the same concentration (Fig. 1C). The pre-sonicated TriP_{96} polymer exhibited no observable reaction with 1-azidopyrene under identical conditions.

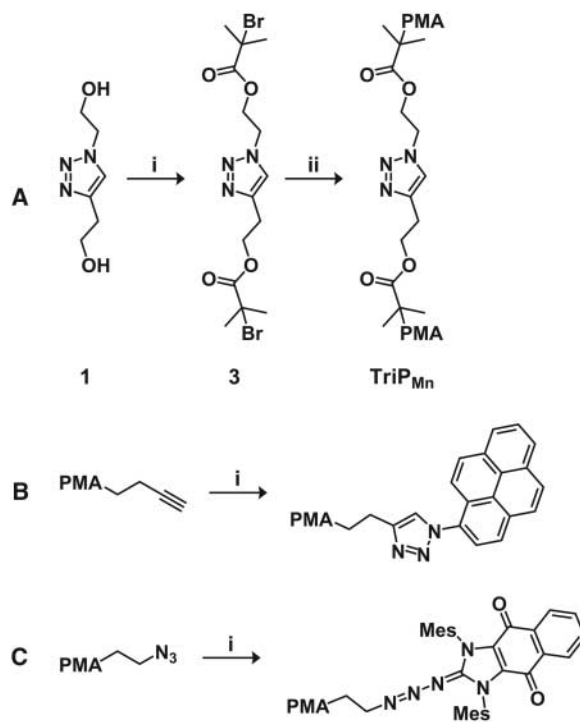
Parallel efforts to selectively label the liberated azide focused on N-heterocyclic carbenes (NHCs), such as 1,3-dimesitylnaphthoquinimidazolyldiene (NQMes), which are known (23) to produce intensely colored, linear diazo amino-imidazoles (DAAs) upon reaction with azides (Scheme 2). As such, NQMes was added to a solution of postsonicated TriP_{96} in dry tetrahydrofuran (THF) under an inert atmosphere. Within 20 min of the NQMes addition, the color of the solution changed from green to blue. UV-vis spectroscopic analysis of the labeled polymer product revealed increased absorbances at $\lambda = 264, 295,$ and 310 nm relative to the starting material, a result that was subsequently confirmed by GPC (21). Collectively, these data are consistent with DAAI formation from an azide-terminated polymer (Fig. 1D). In addition, the pre-sonicated TriP_{96} polymer did not exhibit DAAI formation under identical conditions, as determined by UV-vis spectroscopy and GPC. Thus, the selective labeling of both azide and alkyne functionalities in postsonicated TriP_{96} , but not pre-sonicated TriP_{96} , provided further support for the proposed mechanically facilitated [3+2] cycloreversion process.

Thermally induced cycloreversion was excluded through a series of control experiments. Previous reports (12–18) have demonstrated that sonochemically induced transformations are dependent on the length of the polymer chains attached to the mechanophore (a phenomenon that is not expected if mechanophore activation is



Scheme 1. The application of ultrasound to a triazole embedded within a poly(methyl acrylate) chain results in a formally retro [3+2] cycloaddition (red). The liberated azide and alkyne moieties can be subsequently clicked back together to afford the triazole-based starting material (black).

Scheme 2. (A) Synthesis of triazole containing polymers. (i) 2-bromoisobutyl bromide, triethylamine, ethyl acetate, $0^\circ\text{C} \rightarrow 25^\circ\text{C}$; (ii) methyl acrylate, tris[2-(dimethylamino)ethyl] amine, dimethyl sulfide. PMA, poly(methyl acrylate). (B) Labeling of alkynes generated under sonication with 1-azidopyrene. (i) 1-azidopyrene, CuI, CH_3CN . (C) Labeling of azides generated under sonication with 1,3-dimesitylnaphthoquinimidazolyldiene (NQMes). (i) NQMes, tetrahydrofuran.



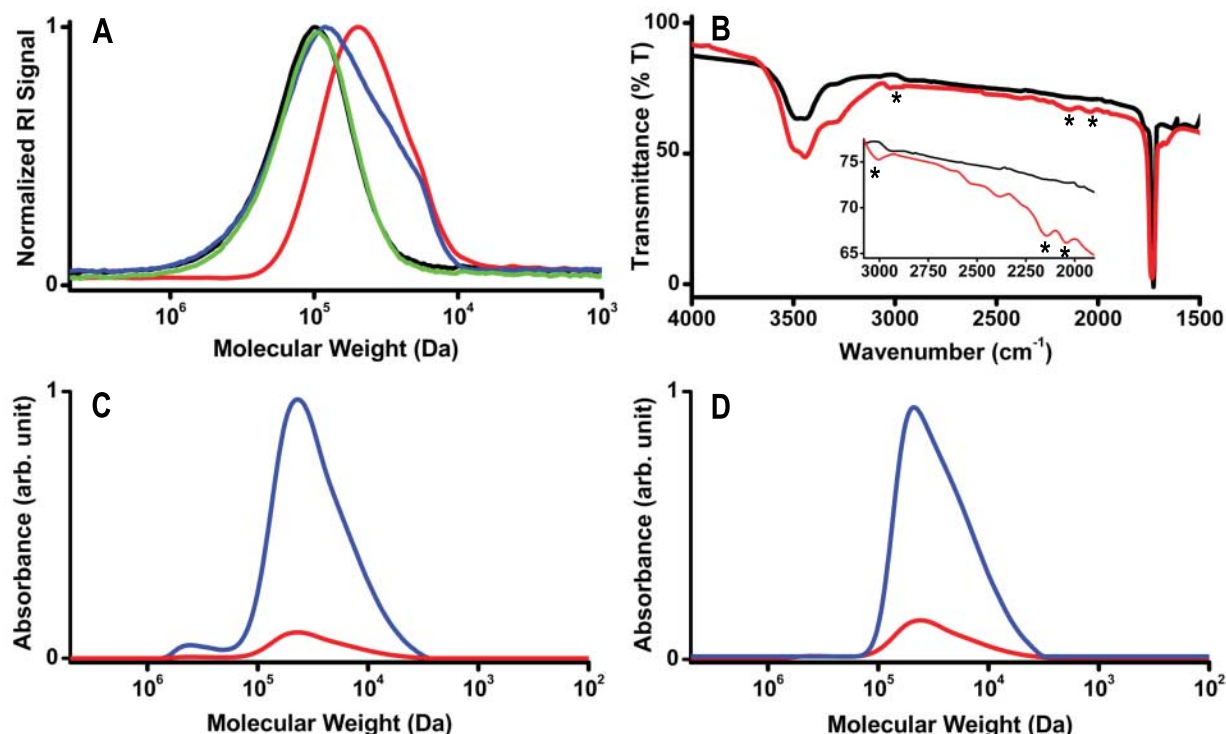


Fig. 1. (A) GPC traces showing the scission of TriP₉₆ (black) under sonication (red) and reformation of the triazole through a CuI-mediated click reaction (blue). Refluxing TriP₉₆ in diphenyl ether (258°C) for 19 hours did not result in a reduction of polymer molecular weight (green). (B) Infrared spectra (in THF) of TriP₉₆ before (black) and after (red) sonication, showing the appearance of the expected azide (N=N=N) and alkyne (CC and CH) stretching frequencies (*) at 2133 cm⁻¹ and 2039/3021 cm⁻¹, respectively. The inset displays a magnified spectrum of the region

from 2000 to 3500 cm⁻¹. (C) GPC traces (visualized using UV detection at 350 nm) showing an increased absorbance for the postsonicated TriP₉₆ polymer upon reaction with 1-azidopyrene (blue) compared with the presonicated TriP₉₆ (red) at the same concentration. (D) GPC traces (visualized using UV detection at 310 nm) showing an increased absorbance for the postsonicated TriP₉₆ polymer upon reaction with NQMes (blue) compared with the presonicated TriP₉₆ (red) at the same concentration.

Table 1. Selected molecular weight data. Tri refers to the 1,2,3-triazole moiety and P_{Mn} corresponds to the approximate number average molecular weight (M_n) of the respective polymer in kD. The polydispersity index (PDI) was calculated using the equation $PDI = M_w/M_n$, where M_w is the weight average molecular weight. M_n and M_w were determined as their polystyrene equivalents by GPC (eluent, THF).

	Presonication		Postsonication		Sonication time
Name	M_n (kD)	PDI	M_n (kD)	PDI	(hours)
TriP ₁₆₀	160	1.2	84	1.3	2
TriP ₉₆	96	1.3	48	1.4	2
TriP ₆₃	63	1.3	32	1.4	5
TriP ₃₆	36	1.3	33	1.3	7
TriP ₁₆	16	1.4	16	1.4	7

a purely thermal process). As such, a series of poly(methyl acrylate) chains of varying M_n were synthesized from **3** via SET LRP (M_n = 16 to 160 kD) (Table 1). Additionally, a small molecule analog of TriP₉₆ featuring pivalate groups in lieu of polymer chains (**2**) was also prepared. The various TriP_{Mn} polymers and **2** were then individually subjected to the sonication conditions described above for TriP₉₆. After sonication, GPC analysis of the isolated materials revealed that chain scission was dependent on the M_n since only higher molecular weight polymers (>36 kD) underwent activation (Table 1). Furthermore, there was no observable alteration of **2** under sonication, as determined by ¹H nuclear magnetic resonance

spectroscopy (21). Thermogravimetric analysis of **2** showed that this compound began to decompose at ~255°C in the solid state; however, when TriP₉₆ was heated in diphenyl ether at a similar temperature (258°C) for 19 hours, there was no observable chain scission by GPC (Fig. 1A, green). Together, these data exclude a thermally promoted cycloreversion of these triazoles as a viable pathway for the chain scission observed under ultrasound.

Finally, we reasoned that if the cycloreversion was mechanical in nature, then the position of the mechanophore within the polymer chain should effect the activation process (12). To explore, triazoles bearing a single polymer chain at either

the 1 or 4 position were synthesized. The resulting materials were subjected to the sonication and labeling experiments as described for TriP₉₆; neither control revealed a reduction in M_n or a positive azide or alkyne labeling result. Collectively, these data suggested that the triazole must be embedded within the polymer chain near the midpoint to experience the force necessary to facilitate cycloreversion (21).

Having established that the observed 1,3-dipolar cycloreversion generated polymer chains terminated with reactive azide and alkyne moieties, we envisioned reclicking the two polymer chains together. TriP₉₆ was subjected to the ultrasonication conditions described above and then analyzed by GPC (M_n = 48 kD). After dissolution in degassed CH₃CN and the addition of CuI, the resulting mixture was stirred at 80°C for 19 hours. Upon isolation of the product, GPC analysis (Fig. 1A, blue) indicated the formation of a polymeric material having a peak molecular weight (M_p , determined by the molecular weight at the peak retention time) similar to that of the presonicated material (i.e., M_p = 90 kD for the coupled product versus M_p = 99 kD for the presonicated material). Deconvolution and integration of the GPC curves obtained upon analysis of this reclicked polymer indicated that the recoupling efficiency was ~86% (21). To

demonstrate the reproducibility of these highly efficient processes, TriP₁₆₀ was successfully subjected to two cycles of mechanically induced cycloreversion and subsequent Cu catalyzed cycloaddition with similar results (21).

Collectively, these results demonstrate that directed mechanical forces can effect the previously elusive retroclick reaction, transforming triazoles into their azide and alkyne precursors. The ability to selectively deconstruct triazoles with high fidelity presents the opportunity to employ these readily accessible moieties in mechanoresponsive materials or as mechanically labile protecting groups. The former could include dye-sensitized force sensors (24) or force responsive fluorescent tags for biological assays. The latter would be of particular appeal for preventing undesired side reactivity in the presence of strong nucleophiles, especially considering that a broad range of substituents covalently bonded to the azide and alkyne coupling partners should tolerate ultrasonication. In a broader perspective, the methodology for triazole cycloreversion established here could be used to coax other kinetically stable molecules to exhibit uncommon

reactivity under the influence of exogenous mechanical forces.

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Supporting Online Material

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Primary Alcohols from Terminal Olefins: Formal Anti-Markovnikov Hydration via Triple Relay Catalysis

Guangbin Dong,¹ Peili Teo,^{1,3} Zachary K. Wickens,¹ Robert H. Grubbs^{1,2*}

Alcohol synthesis is critical to the chemical and pharmaceutical industries. The addition of water across olefins to form primary alcohols (anti-Markovnikov olefin hydration) would be a broadly useful reaction but has largely proven elusive; an indirect hydroboration/oxidation sequence requiring stoichiometric borane and oxidant is currently the most practical methodology. Here, we report a more direct approach with the use of a triple relay catalysis system that couples palladium-catalyzed oxidation, acid-catalyzed hydrolysis, and ruthenium-catalyzed reduction cycles. Aryl-substituted terminal olefins are converted to primary alcohols by net reaction with water in good yield and excellent regioselectivity.

Olefin hydration, the addition of water across a carbon–carbon double bond, is an important industrial process for the synthesis of alcohols (1) and can be readily catalyzed by acids, metal oxides, zeolites, and clays (2). However, in accord with Markovnikov's rule, the proton bonds to the less substituted carbon in these processes (3), and thus, primary alcohols (except ethanol) are difficult to obtain (Fig. 1, equation 1). Given the broad usefulness of pri-

mary alcohols in bulk/fine chemical and pharmaceutical industries (1), there is a compelling need to develop selective catalysts for direct anti-Markovnikov hydration of alkenes (4, 5). Currently, a popular indirect protocol involving hydroboration/oxidation (3) affords hydration products with anti-Markovnikov regioselectivity (Fig. 1, equation 2). This two-step process requires a stoichiometric amount of borane reagents and generates boron waste that is difficult to recycle. Moreover, the peroxides used in the oxidation step raise safety concerns for large-scale production. A hydroformylation/reduction sequence can also produce primary alcohols albeit through a homologation process (Fig. 1, equation 3) (6, 7). Campbell *et al.* recently developed an interesting strategy using a Pd-catalyzed allylic oxidation/ester hydrolysis/olefin reduction sequence to achieve the transformation of termi-

nal olefins to primary alcohols (8). Although effective, this approach requires a three-step operation.

In contrast to anti-Markovnikov olefin hydroamination (9) and alkyne hydration (10), two closely related reactions, very limited success has been achieved toward anti-Markovnikov olefin hydration (Fig. 1, equation 4) (5, 9). In 1986, it was reported that *trans*-PtHCl(PMe₃)₂ (Me, methyl) was able to catalyze hydration of 1-hexene to 1-hexanol (11); unfortunately, this work was difficult to reproduce (5, 12), as was the method in a later report (13). Reliable one-step catalytic protocols are currently limited to certain activated classes of olefin (14, 15). Here, we describe an effective method for the direct synthesis of primary alcohols from nonactivated terminal olefins (defined as non-Michael reaction acceptors) using a triple relay catalysis (16) system, as progress toward an ideal system for catalytic anti-Markovnikov olefin hydration.

Our strategy for anti-Markovnikov olefin hydration is based on a two-catalyst cooperative system involving an oxidation cycle followed by a reduction cycle (Fig. 2). Studies of the Wacker oxidation (17) have established that Pd^{II} salts such as PdCl₂ (X = Cl) can oxidize olefins in the presence of water to produce a carbonyl compound, an acid (HCl), and a palladium hydride (H-Pd-Cl). Precedents also exist for certain metal hydrides (M-H; M = Ru, Ir, Fe, Ni, Pt, etc.) to readily reduce carbonyl compounds, while in the process forming metal alkoxides that can be subsequently protonated by an acid (HX) to give an alcohol and metal salts (M-X) (18). In principle, these two known processes could be combined through a hydride transfer from Pd to M, in which the carbonyl compound and the acid generated from the oxidation cycle constitute the reactants

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Post date 27 June 2014

EDITORIAL EXPRESSION OF CONCERN

In the 16 September 2011 issue, *Science* published the Report “Unclicking the click: Mechanically facilitated 1,3-dipolar cycloreversions” by J. N. Brantley *et al.* (1).

After concerns were raised in an e-mail to the editors from a reader, the corresponding author supervised a comprehensive evaluation of all data presented in the original manuscript by tracing all figures back to their raw data files. In over 50% of the figure parts, the authors deemed the data unreliable due to uncertainty regarding the origin of data or the manner in which the data were processed. A confidential investigation that is relevant to these concerns is currently being conducted by the University of Texas at Austin.

Pending the conclusion of the investigation, *Science* is publishing this Editorial Expression of Concern to alert our readers to the fact that serious questions have been raised about the validity of findings in the Brantley *et al.* paper.

— MARCIA MCNUTT
Editor-in-Chief

Reference

1. J. N. Brantley, K. M. Wiggins, C. W. Bielawski, *Science* **333**, 1606 (2011).

RETRACTION

Post date 20 February 2015

On 27 June 2014, *Science* published an Editorial Expression of Concern about the Report "Unclicking the click: Mechanically facilitated 1,3-dipolar cycloreversions" by J. N. Brantley *et al.* After concerns were raised in an e-mail to the editors from a reader, the corresponding author supervised a comprehensive evaluation of all data presented in the original manuscript by tracing all figures back to their raw data files. In over 50% of the figure parts, the authors deemed the data unreliable due to uncertainty regarding the origin of data or the manner in which the data were processed. The University of Texas at Austin conducted a confidential investigation and shared the conclusion that scientific misconduct had occurred, but provided no further detail of the nature of the misconduct. After the conclusion of the investigation, authors Bielawski and Brantley volunteered to withdraw the paper; it has not been possible to contact author Wiggins. *Science* is therefore retracting the paper.

MARCIA MCNUTT
Editor-in-Chief

Reference

1. J. N. Brantley, K. M. Wiggins, C. W. Bielawski, *Science* **333**, 1606 (2011).



RETRACTED: Unclicking the Click: Mechanically Facilitated 1,3-Dipolar Cycloreversions

Johnathan N. Brantley, Kelly M. Wiggins, and Christopher W. Bielawski

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